SECTION 7: STANDARD MEDICAL PROCEDURES

7.7 Intraosseous (IO) Access

Indication:

- Requirement to give fluid and unable to obtain IV access.

Contraindications:

- Fracture of the bone selected for IO infusion (Select an alternate site).
- Infection at the site selected for insertion (Select an alternate site).
- Excessive tissue (Severe obesity and/or absence of adequate anatomical landmarks), consider alternate site.
- Osteoporosis (Not a contraindication with a drill).
- Previous, significant orthopedic procedure at the site, prosthetic limb or joint.
- IO access (or attempted) in targeted bone within past 48 hrs.

Considerations:

- Ensure the administration of a rapid and vigorous 10 ml flush with normal saline prior to infusion "**NO FLUSH = NO FLOW"**:
  - Repeat syringe bolus (flush) as needed.
- FAST 1™ IO and T.A.L.O.N.™ IO can be left in place for up to 24 hrs.
- IO through burns:
  - An IO device can be inserted through burned skin as long as the underlying bone has not been compromised. It is important to be aware that swelling due to post-burn edema may be severe enough to affect the stability of the IO access site. **Important**: Reassess site frequently for signs of dislodgement.
- CPR and sternal IO:
  - With the FAST 1™ IO, CPR can be perform with IO in place and/or a C-collar installed.
- IO medication administration is the same as IV administration.
- Use in pediatrics:
  - Pediatric patients use proximal tibial insertion site **only**.
  - Compartment Syndrome is a serious complication that can result if a large infiltration and/or extravasation goes undetected.
  - The IO insertion site should be monitored frequently for any signs of infiltration and/or extravasation.

Equipment:

- Appropriate type of intraosseous needle set based on insertion location.
- Alcohol swabs.
- One (1)10 ml syringe with Sterile Saline solution for flush.
- One (1) infusion tube primed with fluid of choice.
- IV line and IV bag primed and ready for use.

7.7 Intraosseous Access continued

TELEFLEX FAST 1™ IO

Description:

- Muscle-powered (not battery-dependent, spring loaded or pneumatic). Actual force will vary depending on patient anatomy.
- ''All-in-one'' IO engineered for automatic depth control (penetrates 6 mm into the manubrium).
- Can be use on adult patients only.

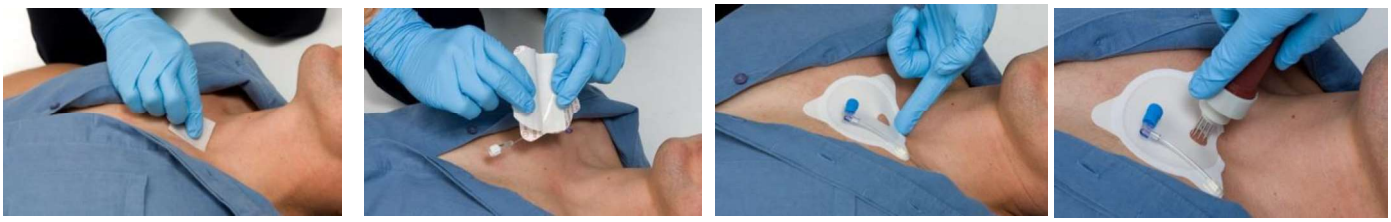
Site of insertion:

- Only sternal.

Procedure:

1. Expose the sternum and locate the sternal notch.
2. Clean infusion site with alcohol swabs.
3. Place Target patch.
4. Stand or kneel at patient head.
5. Twist to remove Sharps Protection Cap.
6. Place Stabilizer needles in target zone.
7. Hold FAST 1™ perpendicular to manubrium.
8. Press down smoothly with increasing force until you hear and feel the Infusion Tube separate from the FAST 1™.
9. Pause & Pull Back: Withdraw FAST 1™ straight back (on axis) while holding down the Target patch.
10. Immediately push Stabilizer needles into the bright red sharps foam plug.
11. Remove blue cap and connect infusion tube to friction fitting on tubing attached to Target patch.
12. Remove white cap from Luer fitting and connect IV tubing.
13. Confirm placement by flushing 2-3 ml into the intraosseous space and draw back into syringe to observe for flashback (blood or marrow). Then flush the contents of the syringe back.
14. Remove the liner from the Protective Dome and apply the Dome over the Target foot infusion site.
15. Begin infusion, secure tubing and monitor insertion site for complications.

7.7 Intraosseous Access continued

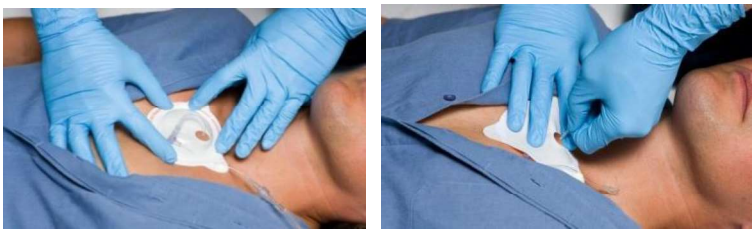


Removal procedure:

1. Remove protective dome from the Target foot.
2. Turn off the source of fluid and disconnect.
3. Grasp infusion tube with fingers as close as possible to patient's skin.
4. Pull perpendicular to manubrium until entire infusion tube emerges from the patient's chest.

Note: Pull in one quick continuous motion until remove.

Note: Use the tube to pull, not the Luer connection. It is normal for the tubing to stretch.

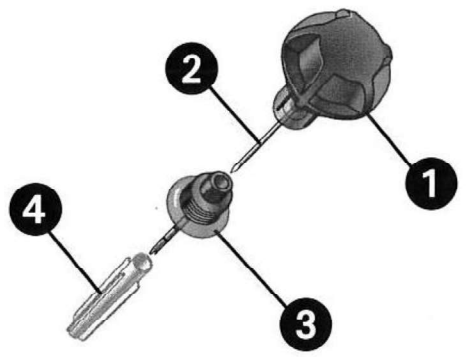


7.7 Intraosseous Access continued

T.A.L.O.N.™

Description:

- Manual Needle Set contain a stylet and a catheter. When the stylet is removed the catheter Luer Lock is exposed. The catheter is a 15 gauge, 38.5 mm, made of 304 stainless and provided sterile, non-pyrogenic in a sealed kit.



Manual Needle Set:

1. Manual Handle
2. Stylet
3. Catheter
4. Needle Cap

Sites of insertion:

- The T.A.L.O.N.™ device provides intraosseous access in the proximal tibia, distal tibia, and humeral head (proximal humerus) of adult and proximal tibia for pediatric patients, when intravenous access is difficult or impossible to obtain for up to 24 hrs.

Procedure (T.A.L.O.N.™):

1. If the patient is conscious, explain procedure.

2. Locate insertion site.

3. Prepare insertion site.

4. Prepare infusion system.

5. Remove and discard the needle set safety cap.

6. Insert:
   a. Control patient's movement prior and during Needle Set insertion;
   b. Position the needle set at a 90° angle to the bone surface;
   c. Gently pierce the skin with the Needle Set until the Needle Set tip touches the bone;
   d. Ensure at least one black line is visible. If not, select different site;
   e. Penetrate bone cortex by rotating clockwise while applying gentle, steady downward pressure;
   f. Stop insertion process when a desired depth is obtained or catheter hub is flushed with the skin (A ''POP'' can be felt when the space is entered);
   g. Do not rock or bend during needle insertion, maintain 90° angle.

7. Remove stylet from catheter by turning counter-clockwise and immediately dispose of stylet in appropriate biohazard sharps container.

7.7 Intraosseous Access continued

<u>Do Not</u> return used stylet to the EZ-IO kit.

8. Confirm catheter stability. Catheter should be stabilized to prevent dislodgement.

9. Attach primed EZ-Connect® extension set to catheter Luer Lock hub.

<u>Do Not</u> attach a syringe directly to the T.A.L.O.N. ™ catheter hub's Luer Lock.

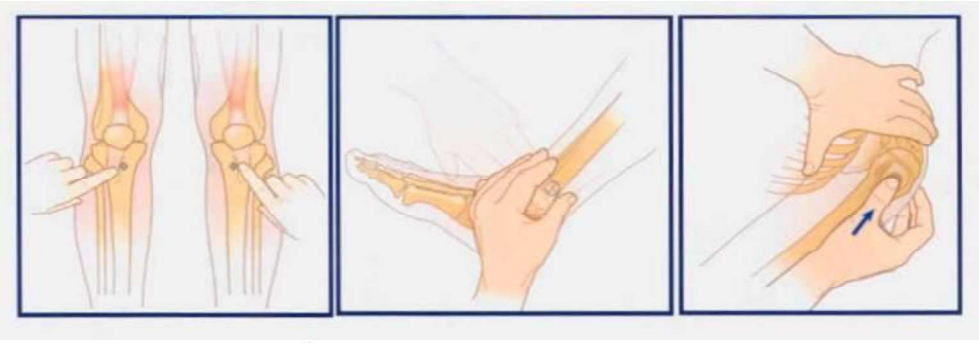
10. Confirm placement by flushing 2-3 ml into the intraosseous space and draw back into syringe to observe for flashback (blood or marrow). Then flush the contents of the syringe back.

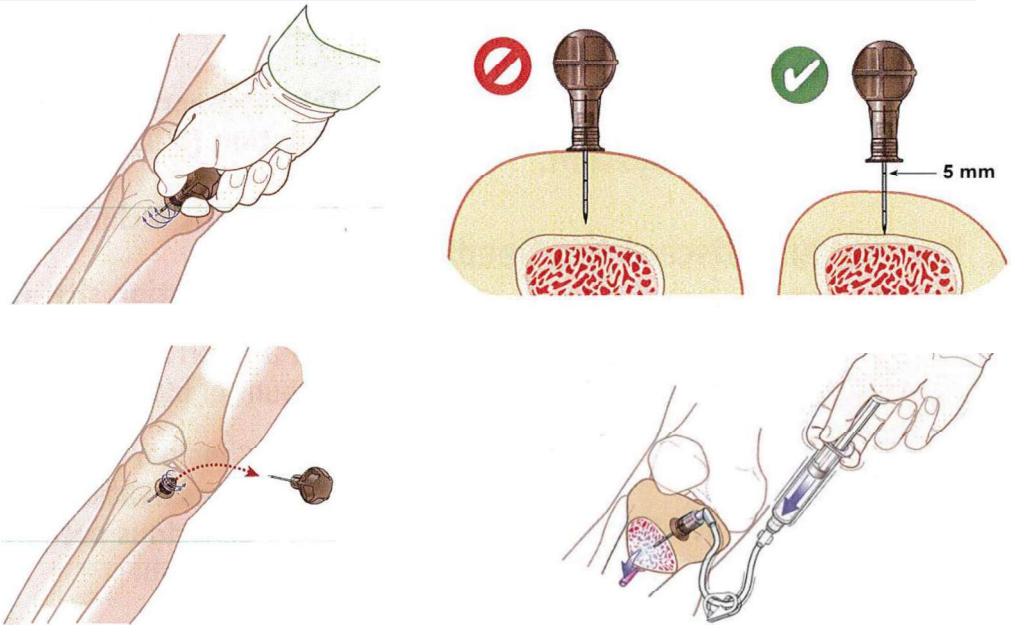
11. Disconnect 10 ml syringe from EZ-Connect® extension set.

12. Connect primed EZ-Connect® extension set to primed IV tubing.

13. Begin infusion, secure tubing and monitor extremity for complications.

Note: Frequently monitor the insertion site for infiltration and /or extravasation.

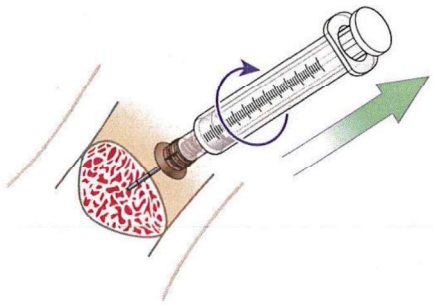




7.7 Intraosseous Access continued

Removal procedure:

1. Attach syringe to the catheter.
2. Gently twist clockwise while slowly applying traction to catheter.
3. Do not rock or bend the catheter during this procedure.
4. Once removed, immediately place catheter in appropriate sharps container.
5. Apply dressing.



Removal procedure